

# SEUPD@CLEF: Team retrix on Source Retrieval for Scientific Web Claims

Fabio Baldan<sup>1</sup>, Giancarlo Padoan<sup>1</sup>, Alvisè Garberino<sup>1</sup>, Marco Tessari<sup>1</sup> and Davide Donati<sup>1</sup>

<sup>1</sup>University of Padua, Italy

## Abstract

We present the results of the retrix group for the CheckThat! 2026 - Task 1, in which the goal is to identify the scientific paper implicitly referenced in a social media post. The input consists of short, noisy and usually informal posts that mention scientific findings without providing explicit links or explicit citations.

Our approach is designed to operate in a multilingual setting, as the dataset includes posts in English, German, and French. To handle this, we used multi-lingual language models.

DA FINIRE CON PIPELINE FINALEEEEEEE DA FINIRE CON PIPELINE FINALEEEEEEE DA FINIRE CON PIPELINE FINALEEEEEEE DA FINIRE CON PIPELINE FINALEEEEEEE

## Keywords

Fact Checking, Information Retrieval, Scientific Document Retrieval, Social media fact verification, CLEF, Neural Re-ranking, Cross Encoder

## 1. Introduction

Introduce the context, motivations, and goals of your project.

The paper is organized as follows: Section 2 describes our approach; Section 3 explains our experimental setup; Section 4 discusses our main findings; finally, Section 5 draws some conclusions and outlooks for future work. To avoid fake news and online misinformation spread through platforms like Twitter, Facebook and Instagram is really important to be able to map the social media claims to a scientific paper that can provide certified and reliable information. This challenge is exemplified by Task 1 of the CLEF 2026 CheckThat! Lab, where participants are tasked with matching each post to the correct scientific paper from a fixed pool of candidate documents.

DA RIVEDERE The proposed solution adopts a two-stage hybrid information retrieval architecture. The first stage generates a candidate list by combining lexical retrieval using BM25, which captures explicit term overlap, with semantic retrieval based on dense embeddings obtained from fine-tuned dual-encoder models (e.g., BGE-m3). FINE REVISIONE

DA VEDERE A FINE PROGETTO The second stage refines the top results by applying neural re-ranking algorithms. Several design questions guide this exploration: which document metadata most usefully inform embedding creation? Does incorporating BM25-mined hard negative examples improve training? And to what degree do cross-encoders—including both task-specific models like SciBERT or MedBERT and more general LLM-based cross-encoders—enhance overall retrieval performance?

Evaluated on the CheckThat! 2026 Task 1 dataset our system ranked XXXXXXXXXXXXXXX on the official leaderboard attaining a Mean Reciprocal Rank at 5 (MRR@5) of XXXXXXXXXXXXXXX% on the development set (XXXXXX place) and XXXXXXXXXXXXXXX% on the test set (XXXXXXXXXX out of XXXXXXXXXXX teams). Key contributions include a robust hybrid retrieval framework, empirical evidence on the effectiveness of embedding fine-tuning and LLM-based re-ranking for bridging informal-to-formal domain gaps.

---

*“Search Engines”, course at the master degree in “Computer Engineering”, Department of Information Engineering, and at the master degree in “Data Science”, Department of Mathematics “Tullio Levi-Civita”, University of Padua, Italy. Academic Year 2025/2026*

✉ fabio.baldan.1@studenti.unipd.it (F. Baldan); giancarlo.padoan@studenti.unipd.it (G. Padoan); alvisè.garberino@studenti.unipd.it (A. Garberino); marco.tessari.8@studenti.unipd.it (M. Tessari); davide.donati@studenti.unipd.it (D. Donati)



© 2026 Copyright for this paper by its authors.

Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).

## 2. Methodology

Describe the methodology you have adopted, the architecture of your system, your workflow, etc.

We started by implementing a basic pipeline using BM25 as baseline, using Apache Lucene as the underlying framework. We then experimented with different stop lists, filters and weightings to improve the MRR@5 score, used as final score in the CheckThat! Lab. We then proceeded to experiment with multi-stage pipelines based on neural models, which is the final structure of our project.

We implemented a three-stage pipeline for our information retrieval system, consisting of:

- **Query expansion:** We expanded the original query with additional context to bridge the gap between informal and scientific language.
- **Retrieval:** We retrieved relevant documents from the corpus using the expanded query.
- **Reranking:** We reranked the retrieved documents based on their relevance to the original query.

For each of these stages, we experimented with different models, choosing the ones that performed better in relation to the others.

### 2.1. Query Expansion

### 2.2. Retrieval

### 2.3. Reranking

### 2.4. Evaluation

## 3. Experimental Setup

Describe the experimental setup, i.e.

- used collections
- evaluation measures
- url to git repository and its organization
- hardware used for experiments
- ...

## 4. Results and Discussion

Provide a summary of the performance on the previous year dataset.

Discuss the results and any relevant issues.

## 5. Conclusions and Future Work

Provide a summary of what are the main achievements and findings.

Discuss future work, e.g. what you may try next and/or how your approach could be further developed.

## 6. Group Members Contribution

**Jane Doe** Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

**Table 1**  
Frequency of Special Characters

| Non-English or Math | Frequency   | Comments          |
|---------------------|-------------|-------------------|
| Ø                   | 1 in 1,000  | For Swedish names |
| $\pi$               | 1 in 5      | Common in math    |
| \$                  | 4 in 5      | Used in business  |
| $\Psi_1^2$          | 1 in 40,000 | Unexplained usage |

**John Doe** Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

**Jane Doe** Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

**John Doe** Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

**Jane Doe** Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

**John Doe** Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

## 7. Misc [TO BE REMOVED]

### 7.1. Tex Files

Put your  $\text{\LaTeX}$  files into the `section` folder as shown in the examples above.

### 7.2. Figures

Put your figures into the `figure` folder and put the caption under the figure. Example of reference to Figure 1.

### 7.3. Tables

Put the caption above the table. Example of reference to Table 1.

See the `booktab` packaged documentation for further options.



**Figure 1:** 1907 Franklin Model D roadster. Photograph by Harris & Ewing, Inc. [Public domain], via Wikimedia Commons. (<https://goo.gl/VLCRBB>).

## 7.4. Bibliography

Example of citations:

- name: ? ]
- parenthesis: [? ]

An initial list of references is provided in the files `bibliography.bib` and `proceedings.bib` that you can expand.

See the `natbib` packaged documentation for further options.

## 7.5. Acronyms

Use the

`\ac{acronym}`

command to insert acronyms, eg. *Average Precision (AP)*. The command will expand the acronym the first time it is used.

An initial list of acronyms is provided in the file `acronyms.tex` that you can expand.

See the `acronym` packaged documentation for further options.